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Highly regio- and stereoselective nitro-oxoamination of mono-substituted allenes†

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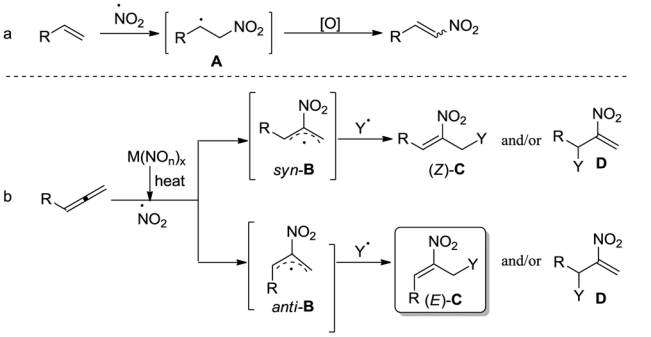
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A highly regio- and stereoselective (up to 99/1) nitro-oxoamination reaction of mono-substituted allene occurs under mild conditions with readily available $AgNO_2$ and a free radical trap TEMPO to form one C–N bond and one C–O bond in one step. Various functional groups and heterocycles could be tolerated in this conversion.

Nitroolefins, highly electron deficient and facilely convertable to other functional groups, have been proven to be useful building blocks[1] and widely used in biological, pharmaceutical and advanced materials science.[2] Therefore, methods for the efficient and highly stereoselective synthesis of nitroolefins are of great interest.[3] The most versatile preparation of nitroolefins involved Henry condensation reaction of aldehydes or ketones with nitroalkanes, which would be followed by dehydration to afford the (*E*)-isomer as the only or major product.[1b,4] On the other hand, different nitrating reagents including $AgNO_2$,[5] $NaNO_2$,[6] $Fe(NO_3)_3 \cdot 9H_2O$,[7] ^{t}BuONO,[8] *etc.*[9] have been explored so far together with a suitable oxidant reagent, such as TEMPO,[10] for the synthesis of nitroalkenes *via* a radical addition process from alkenes (Scheme 1a). Although offering improvements to the direct nitration of alkenes,[5–9] some of these methods also have several drawbacks. Problematic issues including the lack of functional group tolerance and the tendency to form an undesired mixture of *E*/*Z* isomers more or less limit their potential applications. To generate structurally more elaborated and useful compounds, we envisioned the corresponding nitration of mono-substituted allenes, which can be easily prepared from terminal alkynes, formaldehyde, and amines.[11] We expected that the addition of this nitro radical species with mono-substituted allenes should be more favorable due to the formation of a relatively more stable allylic radical intermediate *syn*-**B** or *anti*-**B** (Scheme 1b), which may be directly captured by another radical species, fixing two functional groups in one simple step.[12] Considering the steric effect together with the stability in terms of thermodynamics and kinetics in the radical addition process, intermediate *anti*-**B** shall be favored affording (*E*)-**C** as the major product. To date, there have been only two reports on direct nitration of trialkyl allenes yielding tri- and tetra-substituted nitroalkenes, however, a mixture of *E*/*Z* isomers was formed with poor selectivity (1 : 1–2.8 : 1).[13] Herein, we report the results of highly regio- and stereoselective nitro-oxoamination of mono-substituted allenes[11] with commercially available silver nitrite ($AgNO_2$) and a free radical trap 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO).[10]



Scheme 1 Synthetic methodology of nitroolefins: nitration of alkenes *vs.* nitro-oxoamination of mono-allenes.

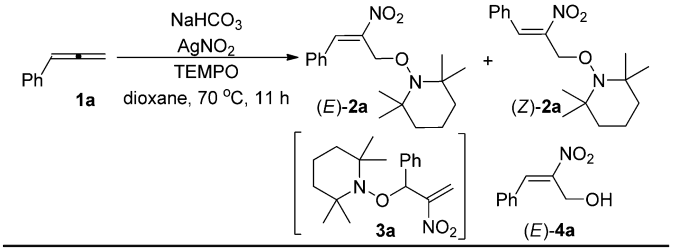
We chose 1-phenylpropa-1,2-diene (**1a**) as the starting material to attempt this conversion with $AgNO_2$ and TEMPO in dioxane at 70 °C. To our delight (*E*)- and (*Z*)-3-(2,2,6,6-tetramethylpiperidinyl-1-oxy)-2-nitro-1-phenylprop-1-ene (**2a**) could be obtained in moderate yield (59%) with a high regio- and stereoselectivity ((*E*)-**2a**/(*Z*)-**2a** = 95/5) as shown in entry 1 of Table 1; **3a** was not detected as judged by the ^{1}H NMR analysis. When $NaHCO_3$ was added as an additive, the yield was improved to 86% (entry 2, Table 1). Although the loading of TEMPO was reduced, $AgNO_2$ and $NaHCO_3$ have a negative effect on the yield, but almost no influence on the stereoselectivity (entries 3–5, Table 1). The reaction is complicated in the absence of TEMPO (entry 6, Table 1).

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† Electronic supplementary information (ESI) available: Detailed experimental procedures and characterization data for all of the new compounds. CCDC 993058 and 993059. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c4cc05743k

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Table 1 Optimization of the nitro-oxoamination reaction of 1-phenylpropa-1,2-diene (**1a**) by varying the dosages of TEMPO, $AgNO_2$ and $NaHCO_3$[a]

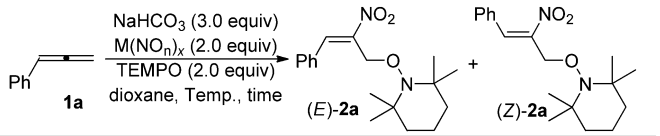


Entry	TEMPO/$AgNO_2$/$NaHCO_3$ (equiv.)	Yield of **2a** (%) (*E*/*Z*)[b]
1	2/2/0	59 (95/5)
2[c]	2/2/3	86 (97/3)
3	1.5/1/1	63 (95/5)
4	1/1.2/1	63 (95/5)
5	1/1/1	69 (97/3)
6	0/2/3	Complicated

[a] The reactions were conducted on a 0.2 mmol scale in dioxane (2 mL). [b] The yield of **2a** and the ratio of (*E*)-**2a**/(*Z*)-**2a** were determined by the ^{1}H NMR analysis with mesitylene as the internal standard. [c] The reaction was conducted at 70 °C for 15 h.

The temperature effect was then explored, and the best results for both yield and regio- and stereoselectivity were obtained at 60 °C (entries 1–5, Table 2). Screening of different nitrating agents, such as $NaNO_2$ (entries 6 and 7, Table 2), $Na_2Co(NO_2)_6$ (entry 8, Table 2), $AgNO_3$ (entry 9, Table 2), $Fe(NO_3)_3{\cdot}9H_2O$ (entry 10, Table 2), and ^{t}BuONO (entry 11, Table 2), did not result in higher yields as compared with $AgNO_2$ (entry 2, Table 2).

Table 2 The effect of temperature and nitration resources for nitro-oxoamination of **1a**[a]

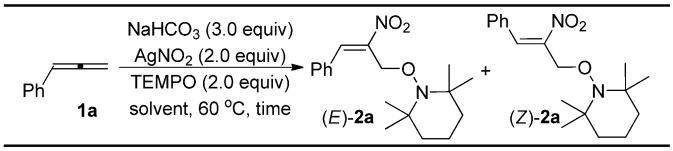


Entry	$M(NO_n)_x$	Temp./time (°C/h)	Yield of **2a** (%) (*E*/*Z*)[b]
1	$AgNO_2$	80/10	65 (93/7)
2	**$AgNO_2$**	**60/11**	**88 (98/2)**
3	$AgNO_2$	50/11	89 (97/3)
4	$AgNO_2$	40/10	70 (97/3)
5	$AgNO_2$	30/10	62[c] (97/3)
6[d]	$NaNO_2$/CAN[e]	60/10	52 (94/6)
7	$NaNO_2$/CAN[e]	60/10	11 (99/1)
8	$Na_2Co(NO_2)_6$[f]	70/34	0[g] (N/A)
9	$AgNO_3$	60/10	0[h] (N/A)
10	$Fe(NO_3)_3{\cdot}9H_2O$[f]	70/34	Complicated (N/A)
11[i]	^{t}BuONO	60/5	70 (95/5)

[a] The reactions were conducted on a 0.2 mmol scale in dioxane (2 mL). [b] The yield of **2a** and the ratio of (*E*)-**2a**/(*Z*)-**2a** were determined by the ^{1}H NMR analysis with mesitylene as the internal standard. [c] With 5% recovery of **1a**. [d] AcOH was used instead of $NaHCO_3$. [e] Cerium ammonium nitrate. [f] The dosages of TEMPO and $M(NO_n)_x$ were 1 and 3 equiv., respectively. [g] With 45% recovery of **1a**. [h] With 56% recovery of **1a**. [i] The reaction was conducted under an air atmosphere.

With the optimized reagents and reaction temperature in hand, we then applied the common solvents for the nitro-oxoamination reaction. Results summarized in Table 3 showed that solvents had a significant effect on the reaction: the reaction could not occur in anhydrous DMF (entry 1, Table 3); a trace amount of (*E*)-**2a** was detected by the ^{1}H NMR analysis when MeCN was used as a solvent (entry 2, Table 3); the reaction in DCE, THF, and toluene could all afford the product (*E*)-**2a** in moderate yields with excellent selectivity (entries 3–5, Table 3). By considering the yields together with the stereoselectivities, the reaction conditions presented in entry 2 of Table 2 have been defined as the standard for further studies.

Table 3 The effect of solvents on nitro-oxoamination of **1a**[a]

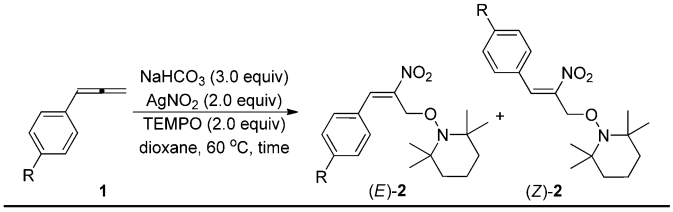


Entry	Solvent	Time (h)	Yield of **2a** (%) (*E*/*Z*)[b]
1	DMF	16	0 (N/A)
2	MeCN	16	3[c] (100/0)
3	DCE	4	60 (96/4)
4	THF	4	69 (99/1)
5	Toluene	4	75 (97/3)

[a] The reaction were conducted on a 0.2 mmol scale in solvent (2 mL). [b] The yield of **2a** and the ratio of (*E*)-**2a**/(*Z*)-**2a** were determined by the ^{1}H NMR analysis with mesitylene as the internal standard. [c] With 7% recovery of **1a**.

Then we explored the scope of the nitro-oxoamination reaction of mono-substituted allenes under the optimized reaction conditions. The substrates with electron-donating groups (methyl, *n*-propyl, and methoxyl groups) (entries 3–5, Table 4), halogen atoms (F, Cl and Br)

Table 4 Highly regio- and stereoselective nitro-oxoamination of 1-arylpropa-1,2-diene[a]



			Isolated yield (%)		
Entry	R	Time (h)	(*E*)-**2**	(*Z*)-**2**	*E*/*Z*[b]
1	H (**1a**)	10	85[c] (**2a**) (98/2)[d]		97/3
2[e]	H (**1a**)	3	91[c] (**2a**) (98/2)[d]		98/2
3	Me (**1b**)	7	43[c,f] (**2b**) (94/6)[d]		96/4
4	Pr (**1c**)	11	88[c] (**2c**) (98/2)[d]		96/4
5	MeO (**1d**)	11	60[c,g] (**2d**) (96/4)[d]		97/3
6	F (**1e**)	6	56 (*E*-**2e**)	—[h] (*Z*-**2e**)	96/4
7	Cl (**1f**)	8	85 (*E*-**2f**)	—[h] (*Z*-**2f**)	95/5
8	Br (**1g**)	12	83 (*E*-**2g**)	—[h] (*Z*-**2g**)	97/3
9	CF_3 (**1h**)	10	56[i] (*E*-**2h**)	—[h] (*Z*-**2h**)	95/5
10	COMe (**1i**)	12	58 (*E*-**2i**)	—[h] (*Z*-**2i**)	97/3
11	CO_2Me (**1j**)	10	86 (*E*-**2j**)	—[h] (*Z*-**2j**)	98/2
12	Ph (**1k**)	7	85 (*E*-**2k**)	—[h] (*Z*-**2k**)	96/4

[a] Conditions: substrate (1.0 mmol), $AgNO_2$ (2.0 mmol), TEMPO (2.0 mmol), $NaHCO_3$ (3.0 mmol), and dioxane (10 mL) at 60 °C. [b] The ratio of (*E*)-**2**/(*Z*)-**2** was determined by the ^{1}H NMR analysis of the crude product. [c] Combined yield of (*E*)-**2** and (*Z*)-**2**. [d] *E*/*Z* ratio after isolation. [e] The reaction was conducted on a 10 mmol scale. [f] The purity of **2b** was 96%. [g] The purity of **2d** was 94%. [h] The *Z*-isomer was contaminated with the *E*-isomer and/or other impurity and could not be isolated in pure form for characterization. [i] The purity of (*E*)-**2h** is 97%.

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(entries 6–8, Table 4), election-withdrawing groups (trifluoromethyl, carbonyl, and ester groups) (entries 9–11, Table 4), or a phenyl group (entry 12, Table 4) at the 4-position of the phenyl group could all undergo this transformation efficiently generating (*E*)-**2** in moderate to good yields and excellent stereoselectivities of (*E*)-**2**/(*Z*)-**2**, respectively. To demonstrate the practicality of the procedure, a gram scale (10 mmol) nitro-oxoamination of **1a** was also conducted in excellent yield (91%) and stereoselectivity (98/2) as shown in entry 2, Table 4. The structure of (*E*)-**2g** was confirmed by the X-ray diffraction studies (Fig. 1).[14]

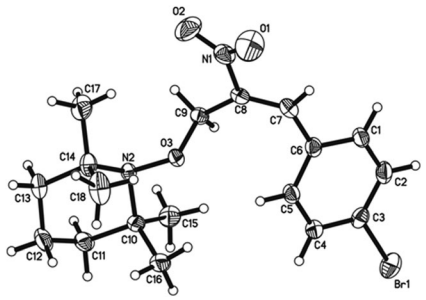
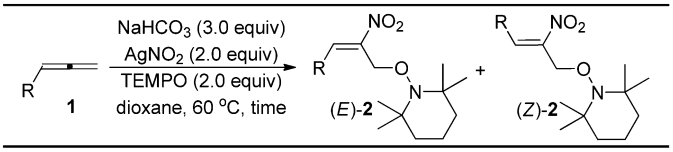
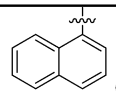
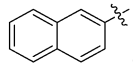
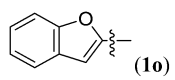
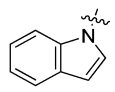


Fig. 1 ORTEP representation of (*E*)-**2g**.

Other 1-aryl substituted mono-allenes were also explored: the reaction of 1-(1- or 2-naphthyl)propa-1,2-diene afforded the expected nitro-oxoamination products smoothly with very high selectivity (entries 1 and 2, Table 5). It is interesting to observe that a great variety of heterocycle based mono-substituted allenes could also be well compatible in this transformation (entries 3–5, Table 5).

Table 5 Highly regio- and stereoselective nitration of mono-substituted allenes[a]



Entry	R	Time (h)	Isolated yield of (*E*)-**2** (%)	*E*/*Z*[b]
1	 (**1l**)	11	85 (**2l**)	99/1
2	 (**1m**)	10	89[c] (**2m**)	97/3
3	 (**1n**)	3	87 (**2n**)	97/3
4	 (**1o**)	6	65 (**2o**)	97/3
5	 (**1p**)	1.5	55 (**2p**)	99/1

[a] Conditions: substrate (1.0 mmol), $AgNO_2$ (2.0 mmol), TEMPO (2.0 mmol), $NaHCO_3$ (3.0 mmol), and dioxane (10 mL) at 60 °C. [b] The ratio of (*E*)-**2**/(*Z*)-**2** was determined by the ^{1}H NMR analysis of the crude product. [c] *E*/*Z* = 98/2 after isolation.

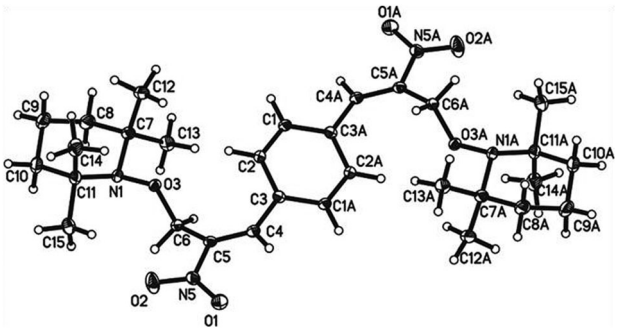
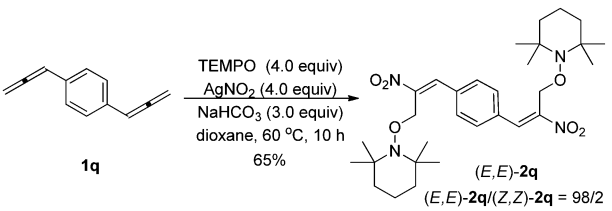


Fig. 2 ORTEP representation of (*E*,*E*)-**2q**.

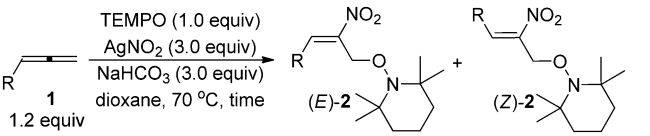
Nitration of 1,4-di(propa-1,2-dien-1-yl)benzene **1q** afforded the double nitro-oxoamination product (*E*,*E*)-**2q** under optimized reaction conditions in moderate yield with excellent selectivity as well (eqn (1)). The structure of (*E*,*E*)-**2q** was confirmed by the X-ray diffraction studies (Fig. 2).[15]

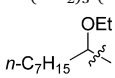
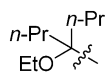
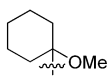


(1)

Furthermore, alka-1,2-dienes could also undergo such a nitro-oxoamination reaction smoothly with high selectivities, albeit with lower yields (Table 6). Besides linear alkyl groups (n-$C_{10}H_{21}$ and $Ph(CH_2)_3$) (entries 1 and 2, Table 6), the other secondary (entry 3, Table 6) and tertiary alcohol ethers (entries 4 and 5, Table 6) are also compatible.

Table 6 Highly regio- and stereoselective nitration of 1-alkylpropa-1,2-diene[a]

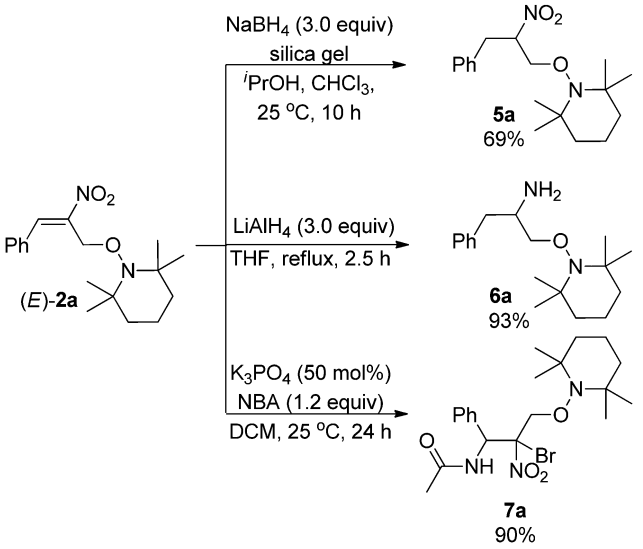


			Isolated yield (%)		
Entry	R	Time (h)	(*E*)-**2**	(*Z*)-**2**	*E*/*Z*[b]
1	n-$C_{10}H_{21}$ (**1r**)	11	29[c] (**2r**)	(97/3)[d]	98/2
2	$Ph(CH_2)_3$ (**1s**)	11	39 (*E*-**2s**)	3 (*Z*-**2s**)[e]	95/5
3	 (**1t**)	11.5	42 (*E*-**2t**)	9 (*Z*-**2t**)[f]	83/17
4	 (**1u**)	11	35[c] (**2u**)	(97/3)[d]	96/4
5	 (**1v**)	12	43[c] (**2v**)	(97/3)[d]	96/4

[a] Conditions: substrate (0.24 mmol), $AgNO_2$ (0.6 mmol), TEMPO (0.2 mmol), $NaHCO_3$ (0.6 mmol), and dioxane (2 mL) at 70 °C. [b] The ratio of (*E*)-**2**/(*Z*)-**2** was determined by the ^{1}H NMR analysis of the crude product. [c] Combined yield of (*E*)-**2** and (*Z*)-**2**. [d] *E*/*Z* ratio after isolation. [e] The purity of (*Z*)-**2s** is 55%. [f] The purity of (*Z*)-**2t** was 83%.

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The nitroolefins prepared here have been demonstrated to be useful in organic synthesis as shown in Scheme 2: (*E*)-**2a** was treated with $NaBH_4$ (3.0 equiv.) and silica gel in a mixture of solvents of ^{i}PrOH and $CHCl_3$ at 25 °C for 10 h to afford 2,2,6,6-tetramethyl-1-(2-nitro-3-phenylpropoxy)piperidine (**5a**) in 69% isolated yield.[16] While amine **6a** could be obtained in 93% yield when $LiAlH_4$ (3.0 equiv.) was used as the reducing agent in refluxing THF for 2.5 h.[17] An addition reaction of (*E*)-**2a** with *N*-bromoacetamide (NBA, 1.2 equiv.) could also be realized in the presence of K_3PO_4 (50 mol%) producing **7a** in 90% yield.[18]



Scheme 2 Synthetic applications of the nitro-oxoamination product (*E*)-**2a**.

In summary, we have developed a highly regio- and stereoselective nitro-oxoamination reaction of mono-allenes *via* a radical process to form C–N and C–O bonds in one step, which could tolerate various functional groups and heterocycles as the substituted groups producing a series of useful building blocks with potential bioactivity.[2] The starting material can be easily available.[11] This conversion would offer a practical strategy for the synthesis of intricate structures. Further studies on nitro-oxoamination of di- and multi-substituted allenes are underway in our laboratory.

Financial support from the National Natural Science Foundation of China (21232006) and the National Basic Research Program (2011CB808700) is greatly appreciated. Shengming Ma is a Qiu Shi Adjunct Professor at Zhejiang University. The results presented in entries 7 and 12 of Table 4 and entry 2 of Table 5 have been kindly reproduced by Mr Tao Cao.

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